

<http://www.freehaven.net>

The Free Haven project: Distributed Anonymous Storage Service

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summary

- problem statement
- terminology
- related work
- freehaven design
 - basic idea
 - publication
 - trading
 - accountability
 - possible attacks
- conclusions

problem statement

- storage system
 - anonymous
 - resists powerful adversaries to find/destroy data
- goals
 - anonymity - publishers, readers, servers
 - persistence - determined by publisher
 - flexibility - add/remove nodes
 - accountability - reputation

terminology

- anonymity
- pseudonymity
- agents: author, publisher, server, reader
- safe communication channel - cypherpunks remailer blocks
- symmetric keys: PK (public) and SK (secret)
 - $PK(SK(doc)) = SK(PK(doc)) = doc$
 - $SK(doc)$ = authenticates sender
 - $PK(doc)$ = authenticates reader
- $hash()$ - noninvertible functions (MD5)

anonymity

- author-anonymity (document $\nrightarrow$ author)
- publisher-anonymity (document $\nrightarrow$ publisher)
- reader-anonymity (document $\nrightarrow$ readers)
- server-anonymity (document $\nrightarrow$ servers)
- document-anonymity (server does not know what documents it stores)
 - passive: cannot recreate document based **only** on data it stores.
 - active: may compare data with other servers.
- query-anonymity (request $\nrightarrow$ document)

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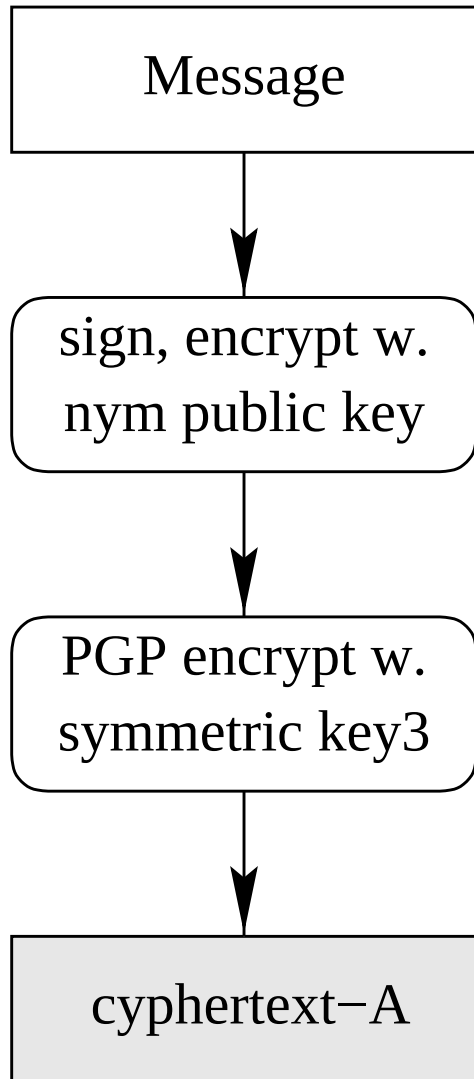
anonymous remailers - cypherpunk

- a network of servers running specialized mail servers
- a message
 - is sent across a chain of remailers
 - is encrypted with *PK* of remailers in the chain
- a remailer
 - only knows identity of prev/next remailer
 - serve **anonymous, not pseudonymous** email
 - can send to unknown destinations

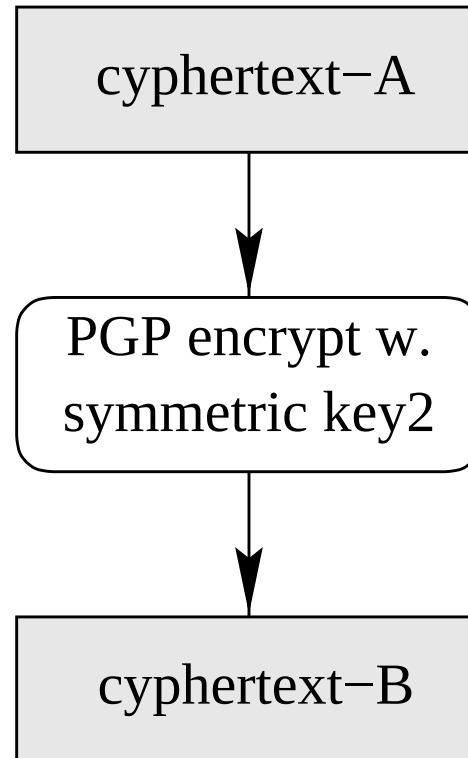
pseudonym remailers - nymserver

- use cypherpunk remailers and a pseudonym server
- to send email from the nym, one sends it to **send@nymserver.net**
- to config the nym, requests go to **config@nymserver.net**
- for each (pseudo)nym, the server keeps:
 - *PK*
 - all mail from the nym is encrypted with the *SK*
 - all mail to the nym is encrypted with this *PK*
 - reply block
 - cypherpunk chain to the true destination
 - the last remailer(innermost) knows the real address
 - the real address may be **alt.anonymous.messages**
 - configuration settings

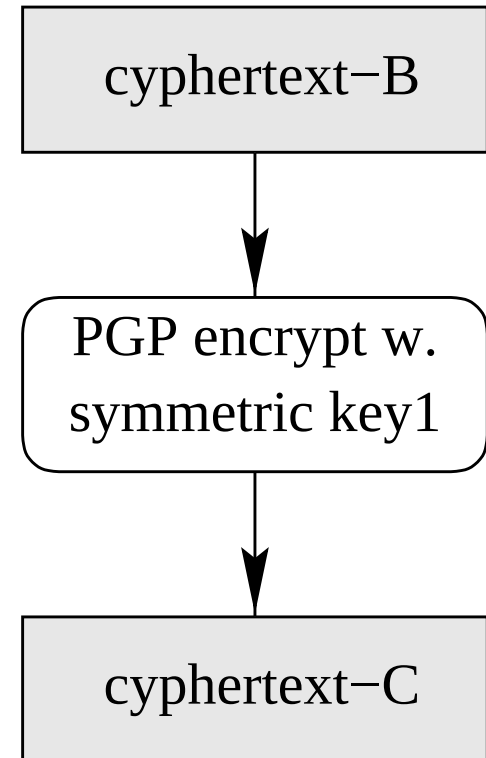
Pseudonym Server:



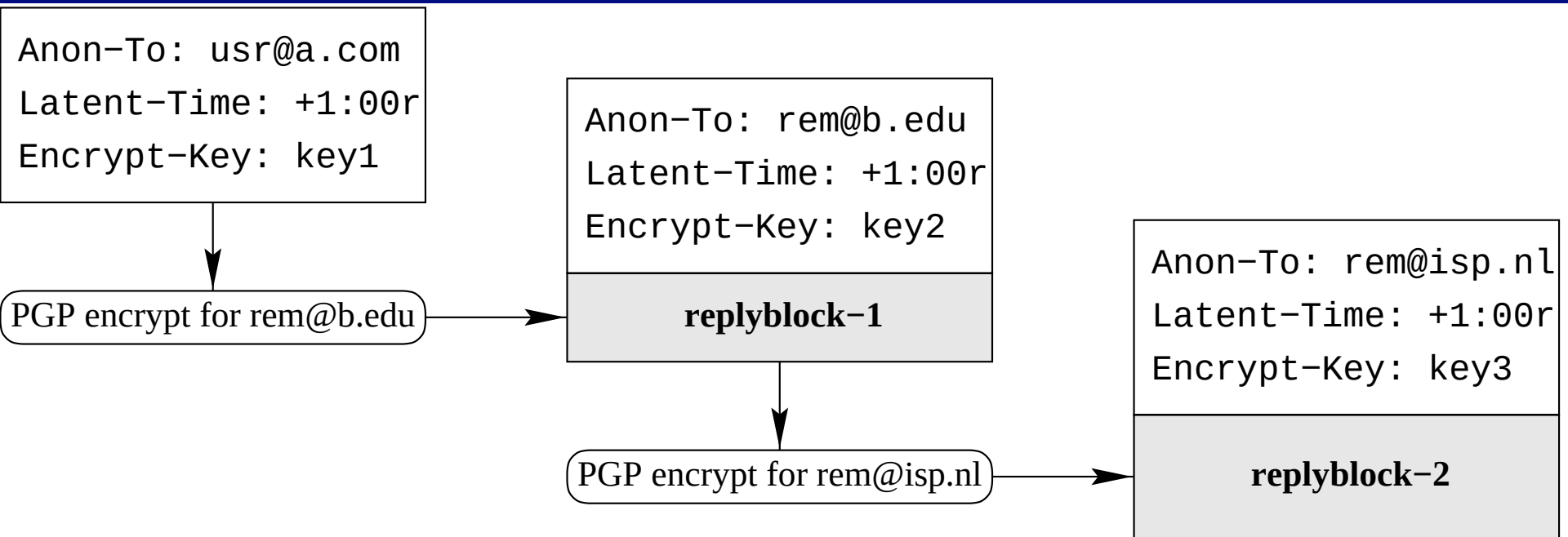
rem@isp.nl:



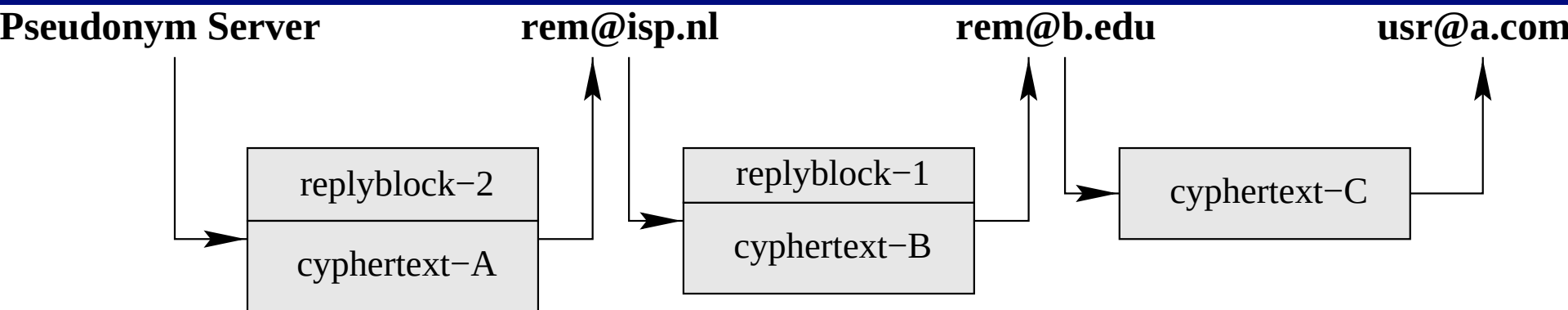
rem@b.edu:



b. Encryptions performed on messages at each remailer



a. Steps performed by a user to construct a reply block with two hops



c. The actual data that traverses the network

related work - napster, gnutella

○ napster

- central directory with filenames
- server supports directory searching
- logical p2p, but physical client-server
- no anonymity

○ gnutella

- true p2p
- dynamic graph - each node chooses its degree
- a query has a TTL and is sent to all neighbors
- nodes directly report hits
- no anonymity - “Gnutella Wall of Shame”
- bottlenecks in the middle
- bootstrap - implicit hierarchy

related work - freenet

- high accessibility
- documents are encrypted with their names
- query
 - hash of the document name
 - fwd to more likely nearby server
- document lifetime $\Leftrightarrow$ popularity
- anonymity: sender, reader, server

related work - mojo nation

- focus is accessibility, **not anonymity**
- digital cash = mojo
- central bank to handle mojo transactions
- trackers:
 - content - keeps sharemaps
 - metatracker - what servers keep what shares
 - publication - tracks documents
- each server has a bitmask describing allowed shares

related work - mojo nation

○ publish

- doc is split in 8 pieces, any 4 of which can rebuild the original
- hashes of the 8 pieces $\Rightarrow$ sharemap
- sharemap is split in 8 $\Rightarrow$ content tracker
- pay to store a share on a server

○ query

- content tracker $\rightarrow$ info about 8 pieces of sharemap
- metatracker $\rightarrow$ servers covering address spaces for the pieces
- buy pieces and rebuild the sharemap
- repeat for the document pieces

○ document lifetime $\Leftrightarrow$ popularity

related work - eternity service

- publisher
 - uploads doc and digital cash
 - uploads to 100 servers, monitors only 10
- servers
 - broadcast queries
 - delivery through anonymous remailers
- problems
 - digital cash
 - what to do with cheating servers (stop payment)
 - no smooth join/leave

related work - USENET “eternity”

- uses normal NNTP for retrieval/posting/expiring
- documents are
 - in html format
 - stored in **alt.anonymous.messages**
 - PGP signed by publishers - pseudonym
 - posted through cypherpunks
- reader anonymity - a user download all today's messages
- append only document system
- not censorable

related work - Publius

- focus on availability and anonymity
- publish
 - encrypt document with key K
 - split K in n shares, any j of which can rebuild K
 - sends $K(doc)$ and a share to n servers
 - “name” of the doc = addresses of the n servers
- query
 - run a local web proxy
 - contact j servers, rebuild K
- doesn't have
 - accountability - DoS with garbage
 - smooth join/leave for servers

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freehaven - basic ideas

“anonymity for anonymous storage”

- servnet = community of servers
- true p2p: all clients/users are servers
- users query the entire servnet at once, not one particular server
- publishers convince a single server to publish
- nobody knows where any server is located (pseudonymity)
- servers form contracts to store each other's material
 - trade
 - reputation
- a server gives up space → gets space on other servers

freehaven - publication

- split doc into n shares, k of which can rebuild the file ($k < n$)
 - large $k \rightarrow$ brittle file
 - small $k \rightarrow$ larger share, more duplication
- generate (SK_{doc}, PK_{doc}) and encrypt each share with SK_{doc}
- store on a server:
 - encrypted share
 - timestamp
 - expiration date
 - $hash(PK_{doc})$

freehaven - retrieval

- documents are indexed by $hash(PK_{doc})$
- reader generates $(PK_{client}, SK_{client})$ and a one-time remailer reply block
- reader broadcasts $hash(PK_{doc})$, PK_{client} and the remailer block
 - to all servers it knows about
 - broadcasts may be queued and bulk sent
- servers holding shares with $hash(PK_{doc})$
 - encode the share with PK_{client}
 - send it using the remailer block

freehaven - share expiration

- absolute date
- “price” of a file = $size \times lifetime$
- Freenet and Mojo nation favor popular documents

freehaven - document revocation

○ could be simple:

- store a *hash(delete_key)* inside each share
- broadcast *delete_key* to have all servers delete the shares

○ but:

- allows new attacks
- decreases consistency (unreached shares)
 - poor channels
 - malicious servers dropping delete requests
- $\exists$ revoke capability $\rightarrow$ incentive for adversaries to pressure/threaten

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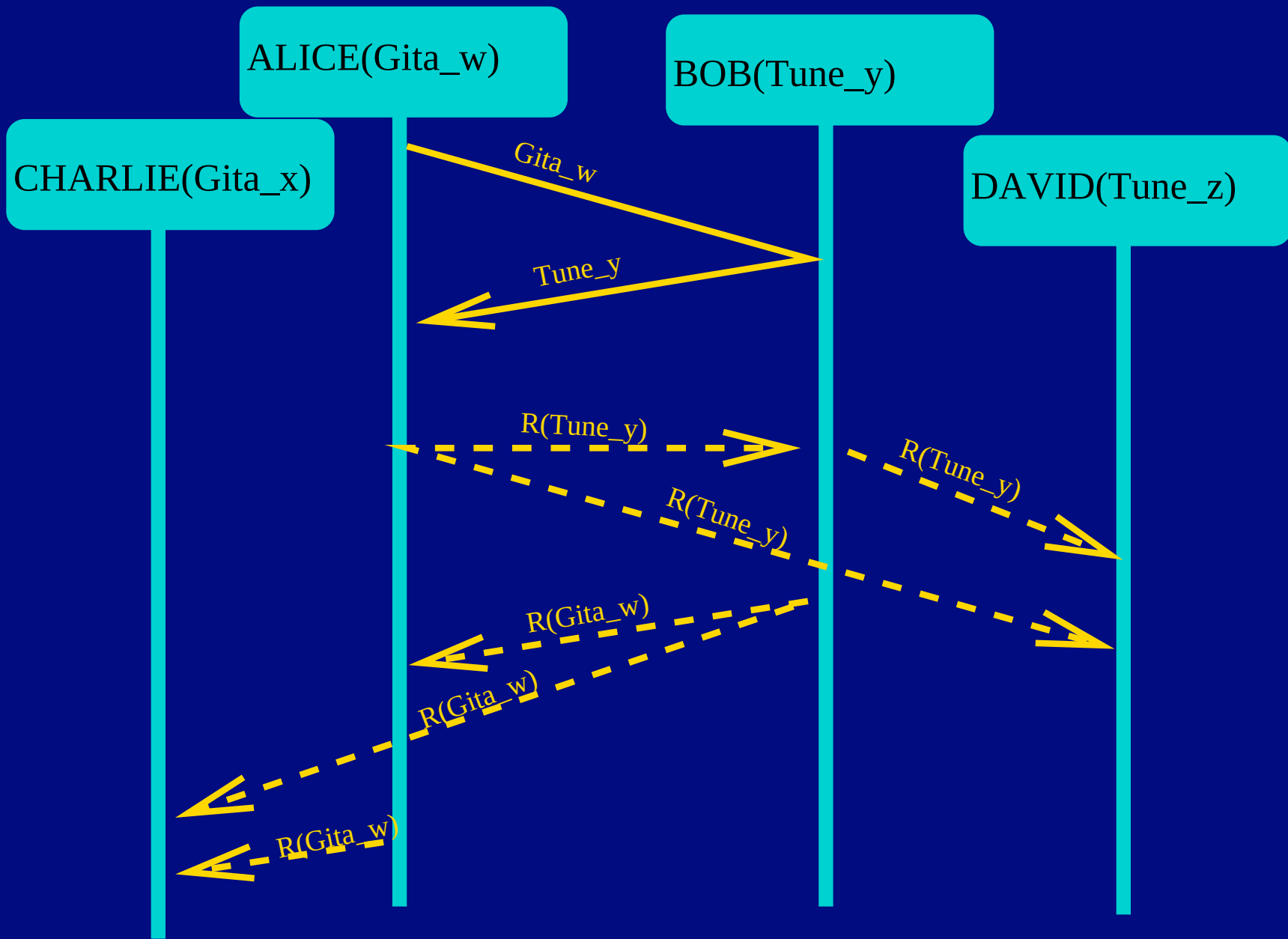
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trading - why?

- to provide a cover for publishing
- to let servers join and leave
 - well behaved, but transient servers
- to permit longer expiration dates
- to accommodate ethical concerns of server operators
 - **attack**: lists of “discouraged” documents
- to provide a moving target

freehaven - trading

- frequency set by server operator
- use reputation to choose partners
- currency is $size \times lifetime$
- rounds
 - exchange shares
 - exchange receipts
 - send receipts to buddies



w and x are buddies; y and z are buddies

freehaven - receipts

- a receipt contains
 - $hash(PK)$ for src/dest servers
 - timestamp
 - foreach share exchanged
 - $hash(PK)$ of the doc
 - share number
 - size, expiration date
- receipt is signed by sender
- complaints are made by showing receipts
- a receipt proves “half” of the transaction

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freehaven - accountability

- each share has a buddy
- buddies know each other's locations
- ?share spawning when buddy disappears?
- when trading, the buddy gets **two** receipts
- what if the buddy moves?
 - servers keep receipts for the lifetime of the traded shares
 - servers forward buddy notifications

freehaven - reputation

- a server keeps track of all other servers
 - reputation: belief that a server will obey the protocol
 - credibility: belief that utterances of a server are valuable
- a server broadcasts referrals
 - after a completing a trade
 - when buddies are lost
 - when reputation/credibility change substantially

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freehaven - possible attacks

“...adversaries are more powerful than anyone ever expects”

- technical, legal, political, social

- attacks on

- availability of documents
- servnet operation
- anonymity
- accountability
- list is incomplete

attacks on documents

- physical attack - need to find all shares
- legal action - depends on the country
 - list of “discouraged” documents?
- social pressure - servers in different jurisdictions
- DoS - real problem, rely on communication channel
- data flooding
- share hoarding - rely on servnet’s size
 - wealthy server flooding with garbage shares?

attacks on reputation system

- simple betrayal - after a corrupt server gains reputation
 - reputation economy makes it unprofitable
- buddy co-opting
 - assume a large servnet/quantity of shares
- false referrals
 - credibility has an associated confidence
- trading receipt games
 - multi server adversaries, bait and switch
- entrapment
 - need a more thorough system of attestation/protest

attacks on anonymity

- most of these are protected by the communication channel - Cypher-punks
- reader anonymity
 - viruses
 - corrupt servers may claim to have a doc and log requests
 - correlate readers and build profiles
- server anonymity
 - create large shares to narrow the possible server set
 - viruses and trojans
- publisher anonymity
 - log publishing acts

anonymity comparison

Project	Publisher C PF	Reader C PF	Server C PF	Document C	Query C
gnutella					
eternity USENET	+ +	?		+	
freenet	+ +			+	
mojonation	? ?			+	
publius	? ?			+	
freehaven	+ +	+ +	+	+why?	

C - computational anonymity

PF - perfect forwarding anonymity

- server pseudonyms can be broken in freehaven
- active servers know what they store

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conclusions

- freehaven = decentralized **anonymous** storage system
 - better for anonymity/persistence than for querying
- unsolved problems - some vulnerabilities:
 - large corrupt servers
 - list of “discouraged” documents
 - DoS
- not ready for wide deployment
 - inefficient communication → few users → weak anonymity

Q & A